

Supplemental Material for “From a Free-Boundary MOTS Saddle-Node to a Spacelike Marginal Tube in Dynamical Five-Dimensional Gravity”

Ronald Bibb

Independent Researcher, Lilburn, Georgia, USA

Correspondence: ronbibb@gmail.com

28 August 2026

1 Purpose and evidentiary scope

This supplement gives the definitions, acceptance criteria, extended numerical results, negative studies, and reproducibility structure underlying the article’s two linked inferences. First, an independently constructed G9/G10/G11 parent sequence evolves from no accepted upper-brane-capped marginally outer trapped surface (MOTS) to two opposite-stability branches, and pseudo-arclength continuation resolves their common critical surface. The principal zero mode, endpoint-eliminated discrete-adjoint coefficients, and invariant one-half area law establish a finite-resolution free-boundary MOTS saddle-node in the $SO(3)$ -symmetric sector. An independent G10 half-timestep evolution repeats the complete critical analysis.

Second, the stable outer branch is followed over a fixed post-fold interval. Its leaves form a future-outward spacelike marginal tube with negative inward expansion, increasing area, positive principal stability, and orthogonal brane contact. The causal character and geometry transfer across G9/G10/G11 and an independent G10 half-timestep evolution. A reflected, brane-inclusive quasi-local charge ledger closes locally and over a finite segment. This supports a finite-resolution free-boundary dynamical-horizon-segment interpretation in the selected symmetry sector and foliation [1, 2].

The saddle-node mechanism itself is established theory [3]; it is not claimed as a new mechanism. The combined result is numerical and quasi-local. It is not an event-horizon reconstruction, a proof excluding every possible initial marginal surface, a continuum theorem, a nonlinear-stability theorem, or a global mass or source-ownership ledger. It also does not determine the nonsymmetric nonprincipal spectrum. Those exclusions are part of the result’s scope, not caveats to be silently relaxed.

The validation record is organized by scientific risk. Each principal test has four elements: the alternative explanation being tested, a criterion fixed before interpretation, the measured outcome, and the consequence for the claim. Operational details—hashes, complete runner histories, environment receipts, and restart records—belong to the reproducibility archive rather than to the scientific argument.

The novelty claim is correspondingly narrow. Cigar-shaped apparent horizons in braneworld initial data and dynamically formed brane-intersecting horizons in RSII precede this work [4, 5],

and MOTS pair creation is already classified as a saddle-node bifurcation [3]. Free-boundary MOTS stability, rigidity, and higher-dimensional structure likewise have an established mathematical literature [6, 7, 8, 9, 10]. Free-boundary minimal annuli also provide a neighboring static fold and Jacobi–Robin degeneration result [11]; that setting does not evaluate the dynamical MOTS adjoint transversality and quadratic coefficients tested here. The unresolved issue addressed here is whether the complete local fold hypotheses are realized when a physical contact condition enters the MOTS boundary-value map and the ambient geometry is supplied by a nonlinear evolution.

The contribution is therefore the closed numerical chain: the physical Robin domain, branch connection, simple principal zero mode, nonzero endpoint-eliminated discrete adjoint projections, invariant square-root separation, three-grid spatial closure, and an independent half-timestep comparison. The stabilized compact braneworld supplies the concrete five-dimensional realization. One such realization establishes that the physical boundary does not obstruct the local fold structure in this problem; it does not establish genericity, a universal theorem, or a new bifurcation mechanism.

2 Model, fields, and conventions

The bulk model is five-dimensional Einstein gravity with $\Lambda_5 = -6/\ell^2$ on the doubled $S^1/\mathbb{Z}_2$ cover. The numerical domain is the fundamental compact interval with fixed branes at its endpoints. The matter sector contains a Goldberger–Wise stabilizer Φ and an independent collapse scalar χ [12, 13, 14, 15]:

$$S = \frac{1}{2\kappa_5^2} \int d^5x \sqrt{-g} (R - 2\Lambda_5) - \sum_i \int_{\Sigma_i} d^4x \sqrt{-h_i} [\lambda_i + U_i(\Phi)] - \frac{1}{2} \int d^5x \sqrt{-g} [(\nabla\Phi)^2 + m_\Phi^2 \Phi^2 + (\nabla\chi)^2 + m_\chi^2 \chi^2], \quad (1)$$

with $U_i(\Phi) = \gamma_i(\Phi - v_i)^2/2$. The induced metric obeys the oriented Israel conditions, the stabilizer obeys the action-derived scalar wall condition, and χ obeys reflecting Neumann data. The computation retains the stabilizer response; it does not integrate out the radion. The displayed action is the doubled-cover thin-shell action. An equivalent one-sided interval variational principle includes the appropriate Gibbons–Hawking–York terms and doubles the one-sided bulk contribution relative to each brane action [16, 17, 18]. With outward normals it yields

$$K_{\mu\nu}^{(B)} = \frac{\kappa_5^2}{2} \left(S_{\mu\nu} - \frac{1}{3} S h_{\mu\nu} \right), \quad n_{\text{out}}^a \nabla_a \Phi = -\frac{1}{2} U'_i(\Phi), \quad (2)$$

which is the sign and factor convention implemented at both interval walls.

We impose $SO(3)$ symmetry in the three noncompact spatial directions while retaining dependence on areal radius r and compact coordinate z . Initial data are momentarily time symmetric,

$$K_{ij} = 0, \quad \Pi_\Phi = 0, \quad \Pi_\chi = 0, \quad (3)$$

and contain a smooth reflection-adjusted scalar pulse localized near the upper brane. The primary datum is the family member labelled $A = 7.90$. The label is a numerical family coordinate, not an invariant critical parameter.

Table 1 gives the dimensionless parameters needed to identify the reported family. We use $\ell = \kappa_5 = 1$, so the compact coordinate spans $z \in [1, e]$. The load-bearing repaired G9/G10/G11 line uses $r/\ell \in [0, 10]$. The superseded pre-repair G7/G8 trajectory, retained only as a historical record below, used $r/\ell \in [0, 8]$. The stabilizer targets are $v_0 = \sqrt{0.01}$ and $v_1 = v_0 e^{-0.1}$, and

Table 1: Primary dimensionless model and discretization parameters.

Quantity	Value
ℓ , κ_5 , compact separation d/ℓ	1, 1, 1
ϵ , Goldberger–Wise backreaction, m_Φ^2	0.1, 0.01, 0.41
wall stiffness $\gamma\ell$	20
wall targets v_0, v_1	0.1, 0.0904837418036
retuned bare tensions λ_0, λ_1	5.9999903768, -6.0528442777
background warp slopes β_0, β_1	1, 1.00880714399
background proper separation/ ℓ	0.9996389584
collapse-scalar mass m_χ^2	0
pulse amplitude A , radial width σ_r	7.90, 1
compact-log width σ_y , center y_c/d , images	0.2, 0.9, 5
historical pre-repair domain and grids	$R_{\max}/\ell = 8$; G7 81×121 , G8 97×145
repaired critical-sequence domain	$R_{\max}/\ell = 10$
repaired critical grids	G9 113×211 , G10 129×241 , G11 145×271
critical and half-step $\Delta t/\ell$	3.125×10^{-5} , 1.5625×10^{-5}
background solve tolerance	10^{-10}

$m_\Phi^2 = 0.1(4.1) = 0.41$. The wall tensions are retuned consistently with the finite-stiffness background solve rather than held at their unstabilized values.

The background values in Table 1 are the sealed binary64 parent values; the last quoted digits of β_1 , the upper-wall tension, and the proper separation vary below 7×10^{-13} , 5×10^{-13} , and 3×10^{-10} , respectively, between the two archived parent resolutions. These numerical values, rather than only their analytic target formulas, are included in the replay package.

For a spatial-slice unit normal u^a and outward surface normal s^a , the outgoing null normal is $\ell_+^a = u^a + s^a$ and

$$\theta_+ = D_i s^i + K_{ij} s^i s^j - K. \quad (4)$$

A MOTS satisfies $\theta_+ = 0$. An upper-brane-capped surface closes regularly on the axis and meets the upper brane orthogonally. Orthogonal contact with the support hypersurface is the free-boundary condition used in the cited literature, and it is precisely the brane endpoint condition imposed here. The cap is therefore a brane-terminating free-boundary surface; free-boundary and capillary MOTS stability have an independent mathematical literature [6, 7, 8, 9, 10], while higher-dimensional MOTSs have also been treated directly [19]. Linear stability is defined by

$$Lf = \delta_{fs} \theta_+. \quad (5)$$

In the convention used here, a nonnegative principal eigenvalue is outward-stable. The evolved-slice operator is treated as non-self-adjoint. The interpretation follows the local MOTS-stability framework of Refs. [20, 21, 22, 23]. For the connected cap, the scalar stability operator is strongly elliptic, and its Robin domain commutes with $SO(3)$. Free-boundary principal-eigenvalue theory supplies a positive eigenfunction that is unique up to scale [6, 10]. A group transform is therefore the same normalized principal eigenfunction. The symmetric computation captures the full principal eigenvalue and its simple kernel, although it does not determine the nonsymmetric nonprincipal spectrum.

The perturbation class is not left implicit. Writing a nearby cap as $\rho(\vartheta) + \delta\rho(\vartheta)$, smooth-axis

closure and preservation of orthogonal wall contact impose

$$\partial_{\vartheta}\delta\rho = 0 \quad \text{at} \quad \vartheta = 0, \pi/2. \quad (6)$$

The operator acts on physical normal displacement $f = w\delta\rho$, where $w = s_A e_{\rho}^A > 0$. Hence the implemented condition is $\partial_{\vartheta}(f/w) = 0$, a homogeneous Robin condition for f when w is nonconstant. The endpoints are eliminated with seven-point local finite differences before the unsymmetrized right-eigenvalue problem is solved. This boundary treatment is checked by the manufactured Robin spectrum, endpoint defects below 10^{-10} , and the 49/65/81-node sequences. No alternative stencil-width study is claimed.

3 Initial-data and search protocol

3.1 Bounded initial search

The load-bearing absence datum is the repaired G9/G10/G11 search at $t/\ell = 0.001$ described in Sec. 9; it is followed by exactly two admitted surfaces at $t/\ell = 0.0015$ on those same parent lines. The earlier searches summarized in this subsection were performed during the superseded pre-repair study and are preserved only to document how the repaired calculation was motivated. They are not independent confirmation of the repaired result. Within that historical record, “no initial MOTS” means “no accepted surface in the prespecified bounded search,” not an analytic nonexistence theorem. The principal initial class consists of C^2 , axisymmetric, star-shaped upper-brane-capped polar graphs

$$z = z_B - \rho(\vartheta) \cos \vartheta, \quad r = \rho(\vartheta) \sin \vartheta, \quad (7)$$

with $0 \leq \vartheta \leq \pi/2$, $\rho'(0) = \rho'(\pi/2) = 0$, and $0.10 \leq \rho(\vartheta) \leq 1.67$. Separate bounded searches cover opposite-side capped, closed-star-shaped, and interval-spanning candidates. Acceptance requires finite geometry, admissible domain and endpoint behavior, a residual below the fixed tolerance, and repeated recovery under the specified seed families.

For the historical $A = 7.90$ datum, the initial upper-brane-capped searches return no accepted surface. The same is true for the five-point R8 sample $A = 7.84, 7.86, 7.88, 7.90, 7.92$. An adverse $A = 7.94$ control recovers its known initial pair. On each separately constraint-solved G9 and G10 initial slice, a regular-axis shooting audit samples 2,049 uniformly spaced axis radii across $[0.10, 1.67]$. The independently minimized brane residual stays positive, 0.0280624085 on G9 and 0.0282193410 on G10, under twelve cutoff, tolerance, and step variations. The same implementation resolves both nearby $A = 7.94$ control roots to residuals of order 10^{-13} . The grade remains a strong floating-point no-root result rather than an interval or degree proof: an arbitrarily narrow unsampled root island is not mathematically excluded.

3.2 Surface representations

On the historical pre-repair trajectory, the primary evolved-surface representation uses the fixed twelve-seed bank 1.15, 1.20, $\dots$, 1.70. Every seed is solved at 40 and 48 cosine modes on 257 collocation points, with expansion target 5×10^{-4} . A candidate requires two-mode convergence in-domain below condition number 10^5 , full profile and endpoint agreement within 0.2%, and an independent 513-point expansion below 0.002; signatures within 0.005 are deduplicated. The confirmation representation is an independently parameterized local boundary-value problem. Acceptance requires agreement in branch count, endpoint geometry, and proper cap area. Both representations operate on the same evolved spacetime, so their agreement constrains detector and parameterization error but is not an independent evolution.

The 10^5 condition-number ceiling belongs only to that historical spectral detector. The load-bearing repaired search and critical continuation below use the distinct local second-order BVP and does not call or inherit that ceiling; its admission uses the local residual, endpoint slope, physical domain, and independent expansion cross-check. Accordingly, proximity to the principal zero cannot remove a continuation point through the spectral-detector condition gate. No critical condition-number trace was archived, and none is used as evidence for the saddle-node classification.

The BVP detector solves the local second-order cap equation with $\rho'(0) = \rho'(\pi/2) = 0$ from the same twelve scalar seeds. Admission requires local interior expansion below 2×10^{-4} , endpoint-slope error below 2×10^{-4} , and a cross-check by the primary expansion evaluator below 0.002. Its flat-slice control reproduces the analytic marginal half-sphere radius 1.25 to 3.91×10^{-11} relatively and gives an independent expansion maximum 3.66×10^{-6} .

In the historical pre-repair record, at $t = 0.000625$, 0.001, and 0.004, each G7/G8 grid yields exactly two accepted branches, with six seed recoveries per branch. The largest shooting/BVP endpoint mismatch is 0.04764% and the largest cross-grid BVP endpoint transfer is 0.23205%.

4 Evolution and ordered acceptance gates

The Einstein–scalar system is evolved in a regular $SO(3)$ -reduced generalized-harmonic formulation [24]. Compact-wall rows own the wall conditions; axis regularity owns interior-axis rows; wall–axis and wall–outer corners use an explicitly staged ownership policy. The canonical policy is the scientific path, while an independent wall-owner policy is retained as a control.

The ordered gate structure prevents a later geometric observation from overriding an earlier invalid computation:

1. runtime, source, and input identity;
2. finite values, signature, and physical admissibility;
3. compact-wall, axis, corner, and ownership identities;
4. repeated deterministic replay;
5. temporal refinement and endpoint consistency;
6. spatial field and physical-tensor closure;
7. horizon count, geometry, and stability interpretation.

An audit or determinism failure is not relabelled as a scientific negative result. A resource stop carries no scientific inference.

5 Formation, geometry, and stability

5.1 Free-boundary stability condition

Let ϱ be the outward unit normal to the brane within the spatial slice, N the unit normal to the MOTS, and ν the outward conormal to its boundary. Orthogonal contact gives $\nu = \varrho$ and $g(N, \varrho) = 0$. For a physical normal variation fN , varying this orthogonality relation gives the free-boundary operator

$$Bf = \nabla_\nu f - \Pi^{\partial M}(N, N)f = 0. \quad (8)$$

Thus the brane-extrinsic-curvature term is part of the physical problem.

For the implemented upper-brane cap, the compact endpoint lies on the upper wall $z = e$. The retained interval lies toward decreasing z , so its outward wall normal is $\varrho = h_{zz}^{-1/2} \partial_z$. We choose the outward cap conormal $\nu = \varrho$, the MOTS normal toward increasing areal radius, $N = h_{rr}^{-1/2} \partial_r$, and $\Pi^{\partial M}(X, Y) = g(\nabla_X \varrho, Y)$. Wall parity gives $h_{zr} = 0$. With these fixed orientations, the physical

normal amplitude associated with a coordinate graph displacement $\delta\rho$ is $f = w\delta\rho$ with $w = \sqrt{h_{rr}}$. The same geometry gives

$$\Pi^{\partial M}(N, N) = \frac{1}{2}\nu(\log h_{rr}) = \nu(\log w), \quad Bf = w\nu(\delta\rho). \quad (9)$$

Consequently the implemented endpoint condition $\partial_\vartheta\delta\rho = 0$, equivalently $\partial_\vartheta(f/w) = 0$, is the coordinate reduction of the physical Robin condition; it does not omit the brane-curvature contribution. The zero derivative at the axis is separately the smooth-axis regularity condition.

5.2 Three-grid formation comparison

The primary repaired critical line is independently constructed on G9 113×211 , G10 129×241 , and G11 145×271 . The prescribed surface search accepts no branch at $t/\ell = 0.001$ and exactly two at $t/\ell = 0.0015$ on every grid. All endpoint, proper-geometry, principal-sign, operator-resolution, and Robin-boundary gates pass. The direct observables are listed in Table 2.

Table 2: Three-grid repaired-parent surface observables at $t/\ell = 0.0015$.

grid	$\mathcal{A}_{\text{in}}$	λ_{in}	$\mathcal{A}_{\text{out}}$	λ_{out}
G9	40.67520320	-0.25894888	40.72845794	+0.29347891
G10	40.67520099	-0.25895908	40.72845823	+0.29348687
G11	40.67519977	-0.25896247	40.72845818	+0.29348984

For G9–G10, the maximum endpoint, geometry, and relative-eigenvalue differences are 1.022×10^{-6} , 1.764×10^{-7} , and 3.938×10^{-5} , respectively. For G10–G11 they are 6.529×10^{-7} , 1.177×10^{-7} , and 1.311×10^{-5} . Thus the repaired line’s earlier absence at $t/\ell = 0.001$ is a pre-formation datum, not a permanent failure to form the pair.

5.3 Pseudo-arclength continuation and zero bracket

Starting from the outer branch at $t/\ell = 0.0015$, time is promoted to an unknown of the local cap BVP and an integral pseudo-arclength condition is added [25]. The spacetime state between exact RK2 endpoints uses the fixed dense extension

$$y(\theta) = y_n + \Delta t[(\theta - \theta^2)k_1 + \theta^2 k_2], \quad \theta = (t - t_n)/\Delta t. \quad (10)$$

Every replayed endpoint is bitwise identical to the stored evolution. Each accepted continuation point starts from arclength step $1/64$ and permits only the prespecified dyadic backoff through $1/4096$; zero-bracket refinement has the technical floor $1/65536$. The augmented BVP uses 121 initial angular nodes, relative tolerance 2×10^{-6} , at most 6,000 adaptive nodes, and a 501-node dense output. An accepted corrector must also have absolute arclength residual at most 2×10^{-5} before the fixed-time surface is re-solved and passed through the inherited local and cross-evaluator gates.

The principal eigenvalue first changes sign and remains resolved under bracket refinement. Table 3 reports the final bracket and near-critical discrete data. The displayed λ_* and coefficients are evaluated at the smaller-magnitude endpoint of the final bracket rather than at an exact floating-point kernel.

The mode convention is exact. The Robin endpoint rows are first eliminated, so the stability matrix acts on physical normal displacement f at the interior uniform ϑ nodes. Its right eigenvector

Table 3: Three-grid critical-time estimate and near-critical discrete operator and normal-form data. $\mathcal{A}_{\text{nc}}$ is evaluated at the selected bracket endpoint rather than an exact critical surface.

quantity	G9	G10	G11
t_*/ℓ	0.0011725421	0.0011725452	0.0011725359
time-bracket width	1.8426×10^{-7}	1.7177×10^{-7}	1.8667×10^{-7}
$\mathcal{A}_{\text{nc}}$, one-sided	40.70345140	40.70336590	40.70344580
λ_*	-7.3854×10^{-4}	-1.6944×10^{-3}	-7.8666×10^{-4}
next reduced-sector eigenvalue	5.56598	5.56531	5.56595
imposed $f_L^* f_R$	1.0000000	1.0000000	1.0000000
near-critical $a = \langle f_L, \partial_t \theta_+ \rangle$	-20.76513	-20.76565	-20.76493
near-critical $b = \frac{1}{2} \langle f_L, \partial_{f_R}^2 \theta_+ \rangle$	2.80717	2.80410	2.81060

is phased so that the component of largest modulus is positive real and then scaled to $\|f_R\|_\infty = 1$. The left vector is the Euclidean conjugate-transpose eigenvector of that reduced matrix, not a mass-matrix adjoint or a separately discretized formal continuum adjoint. It is scaled afterward to $f_L^* f_R = 1$. Thus the reported unit overlap is a construction identity, not an additional nondegeneracy observation. Before rescaling, the code requires only that the overlap magnitude exceed 10^{-14} ; that value was not archived. The endpoint behavior of f_R is restored through the same Neumann extension that implements the Robin graph condition. The near-critical coefficient magnitudes and individual signs are reproducible under this discrete orientation convention, while their nonvanishing is the normal-form conclusion. An admissible change of eigenvector normalization rescales a and b by finite nonzero factors; it changes their numerical magnitudes but cannot change whether either coefficient vanishes.

The near-critical principal mode is real and sign-definite on each grid. The 65–81-node principal differences lie between 3.59×10^{-5} and 3.78×10^{-5} ; the doubled-step differences are approximately 4.78×10^{-4} . The next mode is more than five units away. Centered step halving changes a by at most 1.56×10^{-9} and b by at most 2.67×10^{-3} without changing either sign. Relative adjacent-grid transfers are at most 3.46×10^{-5} for a and 2.32×10^{-3} for b , far below the prospective 0.20 ceilings. The G9/G10/G11 sequences for t_* , $\mathcal{A}_{\text{nc}}$, a , and b are all nonmonotone. Their full relative spreads are, respectively, 7.87×10^{-6} , 2.10×10^{-6} , 3.46×10^{-5} , and 2.31×10^{-3} . Consequently no real three-level Richardson order is defined and no continuum-extrapolated value or uncertainty is inferred. The evidence is small finite-resolution scatter plus the independent temporal comparison below. The isolated near-critical principal mode and bracketed zero crossing are consistent with a simple principal critical kernel. The roles of that kernel, the transversality coefficient, and the nonzero quadratic term are those of the standard fold normal form [26].

5.4 Invariant square-root test

Both branches are solved at the five prespecified offsets $t = t_* + (1, 2, 4, 8, 16)\Delta t/16$. A linear fit of $(\mathcal{A}_{\text{out}} - \mathcal{A}_{\text{in}})^2$ against time gives the results in Table 4. The fitted critical times agree with the continuation estimates within the frozen temporal criterion.

All nine per-grid critical gates and both adjacent-grid aggregate gates pass.

The fit-sensitivity audit separates the two fits rather than treating their statistics as interchangeable. Holding t_* at the independently obtained continuation midpoint gives free exponents 0.49644, 0.49507, and 0.49518 on G9, G10, and G11. Omitting one offset at a time gives the ranges 0.49361–0.49956, 0.49187–0.49865, and 0.49201–0.49872. Fixing $p = 1/2$ and optimizing only the

Table 4: Invariant proper-area scaling fits. The displayed R^2 is for the linear fit of $(\Delta\mathcal{A})^2$ against time; the exponent comes from a separate log-log fit using the critical time inferred by that linear fit.

grid	$t_{*,\text{fit}}/\ell$	free exponent p	squared-fit R^2
G9	0.0011725485	0.4959125	0.9999801
G10	0.0011725343	0.4959672	0.9999804
G11	0.0011725265	0.4959590	0.9999803

amplitude gives maximum relative residuals 0.00920, 0.01173, and 0.01153. These tests support the half-power classification while showing that the seventh decimal place of the original free exponent is not an uncertainty estimate.

5.5 Independent G10 half-timestep comparison

The three spatial grids use the same RK2 step and dense extension. A prospectively frozen G10 calculation therefore halved the step from 3.125×10^{-5} to 1.5625×10^{-5} , reconstructed the parent from $t/\ell = 0.001$ to 0.0015 in 32 steps, solved both endpoint anchors anew, replayed the complete half-step trajectory, and repeated every load-bearing continuation and critical calculation. Table 5 gives the result.

Table 5: G10 half-timestep comparison. The t_* entries are zero-bracket midpoints; their difference is not an uncertainty estimate. The $\mathcal{A}_{\text{nc}}$, a , and b differences are relative, with $\mathcal{A}_{\text{nc}}$ evaluated at the selected near-critical endpoint.

quantity	full step	half step	difference
t_*/ℓ	0.001172545169	0.001172544252	9.1732×10^{-10}
$\mathcal{A}_{\text{nc}}$	40.70336590	40.70340030	8.4495×10^{-7}
a	-20.76564697	-20.76535233	1.4189×10^{-5}
b	2.80409588	2.80435486	9.2348×10^{-5}
free exponent p	0.495967	0.498057	—
squared-fit R^2	0.9999804	0.9999951	—
zero-bracket width	1.7177×10^{-7}	1.7918×10^{-7}	—

The full- and half-step zero brackets overlap. Their midpoint shift is 9.1732×10^{-10} , whereas their widths are 1.7177×10^{-7} and 1.7918×10^{-7} ; the shift is not an uncertainty estimate for t_* . The half-step result retains the real sign-definite near-critical principal mode, reduced-sector spectral gap, nonzero coefficient signs, and area-scaling gate. Both independently evolved parents also pass the same finite-value, signature, constraint, wall, axis, and endpoint gates. This repeat is an indirect check against a systematic temporal-evolution error; it is not a standalone constraint-convergence study. The free exponent moves from 0.495967 to 0.498057, toward the expected one-half value. Together with the fit-sensitivity ranges above, that movement is consistent with the residual exponent offset being dominated by temporal-discretization bias rather than a fitting artifact, but two time steps do not define a temporal convergence order. The prospective limits were $\Delta t/64$ for the bracket-midpoint shift, 1% for the area, 20% for each coefficient, and $0.4 \leq p \leq 0.6$ in both runs. Every comparison passes, so the prescribed decision sequence does not call for a quarter-step run.

5.6 Historical pre-repair formation record

The following G7/G8 trajectory predates the canonical parity repair and is retained for provenance rather than counted as independent evidence for the repaired saddle-node. Both historical grids are empty in the bounded initial upper-brane-capped search and then contain two accepted branches in the common bracket

$$0.000500 < t_{\text{form}} \leq 0.000625. \quad (11)$$

The pair remains accepted at $t = 0.001$ and $t = 0.004$. The independent BVP representation reproduces the zero-to-two transition. Figure 1 shows the shared formation bracket and the subsequent growth of the proper-area separation.

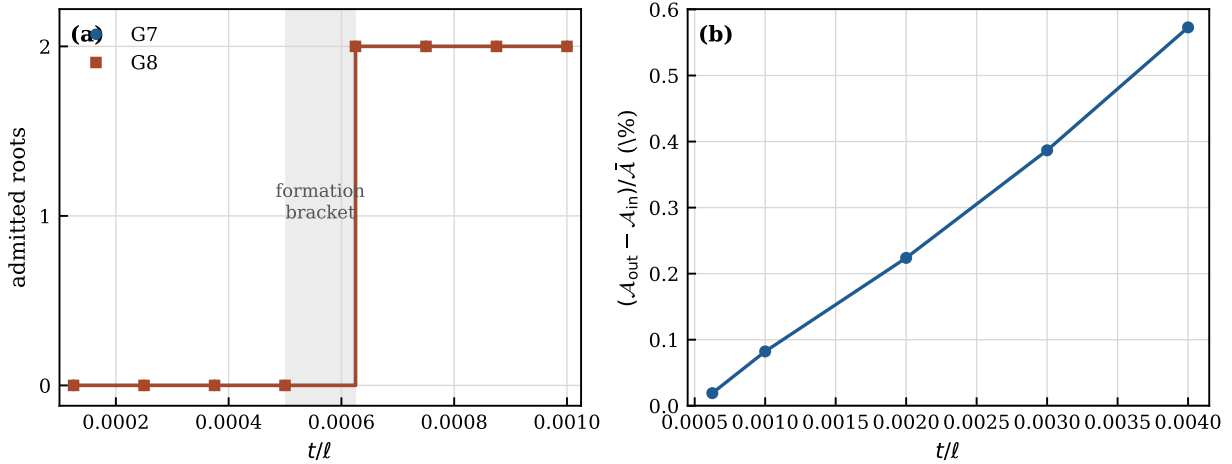


Figure 1: Historical pre-repair realization, retained for provenance and not used as independent support for the repaired result. **Left:** admitted upper-brane-capped root count on the two G7/G8 grids. Both histories change from zero to two in the common formation bracket. **Right:** one-sided proper cap-area separation on G8, shown relative to the mean area at each sampled time. The earliest positive gap is not claimed as quantitatively resolved; the growing separation on later slices is.

5.7 Historical proper-area separation

Selected and confirmation proper cap areas agree within 0.0000881%; the individual G7/G8 proper areas agree within 0.0293%. At first detection the outer-minus-inner gap is 0.02232% on G7 and 0.01908% on G8. Thus both grids reproduce the ordering, while the small gap itself differs by about 16% and is not treated as a precise near-creation observable. Thereafter the inner area decreases and the outer area increases. On G8 the gap grows to 0.5712% at $t = 0.004$, giving a progressively clearer proper-geometric separation.

5.8 Historical principal stability spectrum

The superseded G7/G8 realization also assigned negative principal eigenvalues to the inner branch and positive values to the outer branch at every sampled time. Its full 33/49/65/81-node spectra, Fréchet-step comparison, manufactured control, and numerical values remain in the archive for provenance but are not used to close the repaired result. The load-bearing stability evidence is the

direct G9/G10/G11 result and half-timestep comparison reported above. Neither realization implies nonlinear basin stability.

6 Post-fold marginal-tube construction and controls

6.1 Generator, causal character, and area transport

Let u^a be the future unit normal to a slice and s^a the outward unit normal to a marginal leaf. With

$$\ell^a = \frac{u^a + s^a}{\sqrt{2}}, \quad n^a = \frac{u^a - s^a}{\sqrt{2}}, \quad \ell \cdot n = -1, \quad (12)$$

the normal projection of the positive-time tube generator is

$$X^a = \alpha u^a + v_n s^a = A \ell^a - B n^a, \quad A = \frac{\alpha + v_n}{\sqrt{2}}, \quad B = \frac{v_n - \alpha}{\sqrt{2}}, \quad (13)$$

so that $X^2 = 2AB = v_n^2 - \alpha^2$. A future-outward spacelike tube requires $A > 0$ and $B > 0$; a positive norm alone would also admit the opposite orientation. The reconstruction therefore retains the signs of A , B , and v_n as well as the norm. Backward, centered, and forward velocity paths measure the one-sided temporal spread against which the centered norm is resolved. Equation (13) fixes the normalization and orientation used in every causal-signature comparison below.

The numerical reconstruction starts from the meridional embedding

$$x_i^A(\vartheta) = (-\rho_i \cos \vartheta, \rho_i \sin \vartheta), \quad S_i^A = \partial_\vartheta x_i^A, \quad (14)$$

on leaf i . For physical spacing $\Delta\tau$, the three retained coordinate velocities are

$$\dot{x}_b^A = \frac{x_i^A - x_{i-1}^A}{\Delta\tau}, \quad \dot{x}_c^A = \frac{x_{i+1}^A - x_{i-1}^A}{2\Delta\tau}, \quad \dot{x}_f^A = \frac{x_{i+1}^A - x_i^A}{\Delta\tau}. \quad (15)$$

The corresponding spacetime vectors are $V_p^\mu = (1, \dot{x}_p^A, 0, 0)$, $p \in \{b, c, f\}$. Removing the component tangent to the leaf gives the invariant normal-plane signature

$$\Xi_p = g(V_p, V_p) - \frac{g(V_p, S)^2}{g(S, S)}. \quad (16)$$

With signature $(-, +, +, +, +)$, $\Xi_p > 0$ is spacelike. The centered normal speed in Eq. (13) is reconstructed independently as $v_n = s_A(\beta^A + \dot{x}_c^A)$, where s_A is the unit leaf conormal and β^A the ADM shift.

At each surface node define

$$\delta\Xi = \max(|\Xi_b - \Xi_c|, |\Xi_f - \Xi_c|), \quad m_\Xi = \min_\vartheta (|\Xi_c| - \delta\Xi). \quad (17)$$

A node is resolved only if the three signs agree and $|\Xi_c| > \delta\Xi$. The full/half comparison uses the pointwise normalization

$$\epsilon_{\Xi,p} = \frac{|\Xi_p^{\Delta t} - \Xi_p^{\Delta t/2}|}{\max(|\Xi_p^{\Delta t}|, |\Xi_p^{\Delta t/2}|, 10^{-12})}, \quad (18)$$

and takes its maximum over all paths, nodes, and matched leaves. Equations (14), (15), (16), (17), and (18) therefore define the complete surface-velocity reconstruction and the two normalizations used for the causal classification.

Write the full deformation as $\mathcal{X}^a = X^a + V^a$, with V^a tangent to the leaf. The first variation of the one-sided area is

$$\frac{d\mathcal{A}}{d\lambda} = \int_S (A\theta_{(\ell)} - B\theta_{(n)}) dA + \int_{\partial S} V \cdot \nu dA_2. \quad (19)$$

At orthogonal contact, the leaf conormal ν is the spatial brane normal. Because the endpoint moves within the brane, $\mathcal{X} \cdot \nu = 0$, while $X \cdot \nu = 0$ by normality to the leaf. Hence $V \cdot \nu = 0$. On an exact marginal tube,

$$\frac{d\mathcal{A}}{d\lambda} = - \int_S B\theta_{(n)} dA, \quad (20)$$

so $B > 0$ and $\theta_{(n)} < 0$ imply strict area increase. This is the free-boundary version of the local dynamical-horizon area law [1, 2]. The endpoint cancellation in Eq. (19) and the resulting Eq. (20) distinguish the ordinary area transport from the separate reflected-curvature seam contribution derived next.

6.2 Brane-inclusive reflected charge balance

The ordinary area transport has no separate endpoint term, but the intrinsic curvature balance of the reflected leaf does. Write the one-cap intrinsic metric in proper meridional distance as $dq^2 = ds^2 + W(s)^2 d\Omega_2^2$, set $c = W_s/W$ at the brane, and let $A_2 = 4\pi W_B^2$. The Israel condition fixes the signed seam curvature, whose distributional contribution on the reflected leaf is

$$I_\delta = 8cA_2. \quad (21)$$

Let H_s and $H_\Omega = X \ln W$ be the meridional and spherical principal deformation rates at the joint. Before any balance residual is evaluated, the cap variation, direct seam derivative, and differentiated Israel compatibility condition are fixed as

$$D_{\text{cap}} = 8A_2 [-\partial_s H_\Omega + c(H_s - H_\Omega)], \quad (22)$$

$$\dot{I}_\delta = 8A_2 (\dot{c} + 2cH_\Omega), \quad (23)$$

$$\dot{c} = \partial_s H_\Omega - cH_s. \quad (24)$$

Varying the smooth cap boundary and distribution together therefore combines Eqs. (22), (23), and (24) into the coupled identity

$$D_{\text{cap}} + \dot{I}_\delta = 8A_2 c H_\Omega, \quad (25)$$

where $H_\Omega = X \ln W$. Equation (25) prevents the moving joint from being counted independently as curvature, brane matter, and normal-connection delta terms.

For the reflected three-leaf define

$$\hat{\mathcal{A}} = 2\mathcal{A}, \quad R_A = \left(\frac{\hat{\mathcal{A}}}{2\pi^2} \right)^{1/3}, \quad Q_{\text{AdS}} = \frac{R_A}{4} \left[\int_{\hat{S}} R_{\hat{S}} dA - \frac{2}{3} \int_{\hat{S}} K_\perp^2 dA - \Lambda \hat{\mathcal{A}} \right]. \quad (26)$$

Equation (26) is the reported $\kappa_5^2 E$ convention, with $\Lambda = -6$ and $K_\perp^2 = 0$ on the exact marginal leaf. The intrinsic-curvature integral includes I_δ ; omitting it is a distinct bulk-only comparator, not the reported charge. The coupled seam contribution to the charge rate is

$$F_{\text{seam}}^{\text{coupled}} = \frac{R_A I_\delta}{12} \frac{\dot{\hat{\mathcal{A}}}}{\hat{\mathcal{A}}} + 2R_A A_2 c H_\Omega. \quad (27)$$

No coefficient in Eq. (27) is selected from the final residual.

For completeness, set $h = \dot{\hat{\mathcal{A}}}/\hat{\mathcal{A}}$, $C = h + B\theta_{(n)}$, let ω be the normal-bundle connection along the proper meridional coordinate, and define

$$\mathcal{N}[B, \omega] = B_{ss} + 2\frac{W_s}{W}B_s - 2\omega B_s - B\left(\omega_s + 2\frac{W_s}{W}\omega\right) + B\omega^2. \quad (28)$$

All smooth integrals below are over the reflected cap with the distributional joint removed. The implemented flux inventory is

$$\begin{aligned} F_R &= \frac{R_A}{12} \int R_{\hat{S}} C \, dA, & F_{\ell\ell} &= -\frac{R_A}{3} \int \theta_{(n)} A T_{\ell\ell} \, dA, \\ F_{\ell n} &= -\frac{R_A}{3} \int \theta_{(n)} B T_{\ell n} \, dA, & F_{\sigma} &= -\frac{R_A}{3} \int \theta_{(n)} A \sigma_{(\ell)}^2 \, dA, \\ F_{\omega} &= -\frac{R_A}{3} \int \theta_{(n)} \mathcal{N}[B, \omega] \, dA, & F_{\text{aniso}} &= -\frac{R_A}{2} \int \mathring{R}_{AB} \mathring{H}^{AB} \, dA, \\ F_{\Lambda} &= \frac{\Lambda R_A}{3} \dot{\hat{\mathcal{A}}}, & F_{\text{bg}} &= -\frac{\Lambda R_A}{3} \dot{\hat{\mathcal{A}}}, \\ F_{\text{tang}} &= \frac{R_A}{3} \int \theta_{(n)} \mathcal{L}_X \theta_{(\ell)} \, dA. \end{aligned} \quad (29)$$

Here $\mathring{R}_{AB}$ and $\mathring{H}_{AB}$ are the trace-free intrinsic Ricci and deformation tensors. $T_{\ell\ell}$ and $T_{\ell n}$ are evaluated directly from both evolved scalar fields and the stabilizer potential. The normal-bundle operator in Eq. (28) and the terms in Eq. (29) give the complete rate ledger

$$F_{\text{tot}} = F_R + F_{\ell\ell} + F_{\ell n} + F_{\sigma} + F_{\omega} + F_{\text{aniso}} + F_{\Lambda} + F_{\text{bg}} + F_{\text{tang}} + F_{\text{seam}}^{\text{coupled}}. \quad (30)$$

Equation (30) replaces rather than supplements the older seam bookkeeping. The two seam entries retained in the machine record are the global-radius and joint-work terms displayed separately in Eq. (27); the older curvature, brane-matter, and normal-connection delta entries are replaced, not added. This reflected Hawking–AdS quantity is used only as a quasi-local finite-segment diagnostic; it is not identified with a global or sector-owned mass.

At each leaf the charge-history derivative and all flux terms are evaluated independently. The local residual normalization is

$$\epsilon_{\text{rate}} = \frac{|\dot{Q}_{\text{AdS}}^{\text{FD}} - F_{\text{tot}}|}{\max(|\dot{Q}_{\text{AdS}}^{\text{FD}}|, \sum_i |F_i|, 10^{-12})}. \quad (31)$$

For segment nodes $j = 44, 45, 46, 47$, the composite trapezoid is

$$I_F = \Delta t \left(\frac{1}{2} F_{44} + F_{45} + F_{46} + \frac{1}{2} F_{47} \right), \quad \epsilon_{\text{seg}} = \frac{|\Delta Q_{\text{AdS}} - I_F|}{\max(|\Delta Q_{\text{AdS}}|, |I_{\dot{Q}}|, |I_F|, 10^{-12})}. \quad (32)$$

$I_{\dot{Q}}$ is the same quadrature applied to the independently differenced charge rate. The local and segment normalizations in Eqs. (31) and (32) therefore make the reported 0.30182% a charge–flux comparison, not a term-sum rounding check.

6.3 Spatial, temporal, and balance results

The G10 full lane uses $\Delta t/\ell = 3.125 \times 10^{-5}$ and centered leaves 39–47; the half-step lane uses $\Delta t/\ell = 1.5625 \times 10^{-5}$ and leaves 78, 80, $\dots$, 94. These are nine exactly time-aligned samples over

$0.00121875 \leq t/\ell \leq 0.00146875$, each reconstructed with guard leaves on both sides, so the physical leaf spacing in both comparisons is $3.125 \times 10^{-5}\ell$. The full/half causal comparison includes all three velocity paths and all 501 surface nodes. The G9/G10/G11 geometry and stability transfer uses full-step leaves 43–48; its causal signature uses the four centered interior leaves 44–47. The integrated balance uses those same four nodes over $0.001375 \leq t/\ell \leq 0.00146875$ and spatial stencil widths 5, 7, and 9. Table 6 reports the classification-driving values.

Table 6: Post-fold marginal-tube controls. All stated ceilings were fixed before the corresponding comparison.

control	criterion	outcome
G9/G10/G11 geometry transfer	$< 1\%$	maximum 1.24×10^{-6}
G9/G10/G11 principal stability	$< 10\%$ or 0.02 absolute	signs agree; maximum relative difference 4.47×10^{-5}
resolved causal character	all nodes spacelike	501 of 501 nodes on every tested interior leaf
G10 full/half causal signature	relative difference $< 10^{-2}$	maximum 1.7332×10^{-8}
future trapped signs	$B > 0$, $\theta_{(n)} < 0$	both hold on every accepted leaf
area and stability	$\dot{\mathcal{A}} > 0$, $\lambda_0 > 0$	strict in both G10 timestep lanes
orthogonal contact	bounded residual	maximum 1.10×10^{-18}
integrated charge–flux balance	relative discrepancy $< 1\%$	maximum 0.30182%
adjacent-grid integrated transfer	relative difference $< 1\%$	maximum 7.575×10^{-5}
within-grid stencil transfer	relative difference $< 1\%$	maximum 3.538×10^{-5}

Every matched G10 leaf has negative inward expansion; the largest stored value is approximately -2.8587×10^{-2} . The maximum interior marginality defect is 1.85×10^{-6} . The full/half causal-signature comparison includes all nodes and all three velocity paths. The integrated balance uses a fixed four-node composite trapezoid rule and independently reevaluated charges at the four nodes. On the central width-7 stencil, the charge changes on G9/G10/G11 are respectively 0.05084154, 0.05084091, and 0.05083977; the integrated total fluxes are 0.05099546, 0.05099159, and 0.05099174. The brane-endpoint subtotal is nonzero and supplies approximately 59.44% of the integrated total flux on each grid. This percentage is a local ledger decomposition, not an attribution of global mass or source ownership.

The sign, causal, area, stability, spatial-transfer, temporal-transfer, and balance tests are logically distinct. Together they support a finite-resolution free-boundary dynamical-horizon segment in the selected sector and foliation, rather than only persistent MOTSs on neighboring slices. They do not establish an event horizon, future completeness, or global tube topology.

7 Risk–criterion–outcome–consequence matrix

Table 7 collects the load-bearing validation results in the same risk–criterion–outcome–consequence order used to delimit the claim.

Table 7: Scientific validation matrix.

Risk	Criterion	Outcome	Consequence
Shared temporal-interpolation bias in the critical point	Repeat the complete G10 parent, continuation, and critical analysis at $\Delta t/2$	The $\sim 1.8 \times 10^{-7}$ brackets overlap; their midpoints differ by 9.17×10^{-10} ; relative changes in area, a , and b are below 10^{-4}	Strongly constrains the prescribed temporal concern without treating the midpoint shift as uncertainty; no quarter-step extension is triggered
Unverified direct G11 MOTS observables	Exactly two branches and 1% adjacent-grid geometry transfer at a common time	At $t = 0.0015$ the repaired-parent search admits exactly two opposite-stability branches on G9/G10/G11; maximum geometry transfer 1.76×10^{-7}	Establishes direct three-grid formation geometry and stability on the repaired parent
Boundary discretization changes the critical mode	Repeat angular resolution, Fréchet step, and the physical Robin reduction	The principal sign is stable over 49/65/81 nodes and step doubling; the manufactured free-boundary control passes	Supports the principal mode under the implemented seven-point boundary treatment; no alternative stencil width is claimed
Pair anatomy does not identify a fold	Continue both branches through a principal zero and isolate the near-critical mode	The continuation encloses the zero on all grids; the near-critical mode is sign-definite and the next reduced-sector eigenvalue remains near 5.566	Supports a simple critical kernel rather than only a zero-to-two history
Degenerate or tangential critical point	Nonzero, step-stable near-critical discrete-adjoint coefficients with consistent grid transfer	$a \simeq -20.765$ and $b \simeq 2.808$ retain sign under step halving and across grids	Resolves temporal transversality and quadratic nondegeneracy under the stated discrete convention
Coordinate branch separation	Invariant area difference follows a one-half power with fit sensitivity	Spatial free fits give $p \simeq 0.496$; half-step G10 gives 0.4981; leave-one-out ranges stay 0.4919–0.4996	Supplies an invariant saddle-node scaling test without treating fit digits as uncertainty
Persistent roots do not form a coherent tube	Resolve the normal generator, its orientation, both expansions, area change, and stability on every accepted leaf	Every tested node is spacelike; $B > 0$, $\theta_{(n)} < 0$, $\dot{\mathcal{A}} > 0$, and $\lambda_0 > 0$ persist	Supports a finite-resolution free-boundary dynamical-horizon segment rather than leaf persistence alone

Risk	Criterion	Outcome	Consequence
Tube character is a shared temporal artifact	Repeat the matched G10 leaves on an independently evolved half-timestep parent	The maximum pointwise causal-signature difference is 1.7332×10^{-8} against the 10^{-2} ceiling	Constrains the causal classification under temporal refinement
The open cap omits or double counts boundary flux	Derive the coupled seam term before evaluation and require a sub-1% finite-segment charge-flux discrepancy across grids and stencils	The worst discrepancy is 0.30182%; adjacent-grid and stencil metrics remain below 7.6×10^{-5}	Supports a brane-inclusive local balance without promoting it to a global mass ledger

The complete record also documents evolution gates, discrete eigenvector normalization, the equivariance argument for the principal mode, the absence of an archived condition-number trend, and the nonmonotone three-grid scatter. Those items delimit interpretation but do not add further headline claims.

8 Historical pre-repair studies

The superseded pre-repair G7/G8 trajectory is retained in the archive rather than transferred as validation of the repaired G9/G10/G11 result. It includes fixed 8/16/32-step temporal comparisons, two live-gauge variants, R8/R10/R12 domain checks, a sparse $A = 7.84\text{--}7.92$ amplitude sample, and 480 sealed searches for opposite-side, closed-star-shaped, and interval-spanning candidates. These studies support only their original bounded dispositions: they neither establish repaired-line gauge robustness, a critical amplitude, an open basin, nor a global topology theorem. The load-bearing temporal and spatial tests are instead those reported in Table 5 and the direct three-grid analyses above.

9 Earlier G11 field closure and pre-formation search

Before the direct formation calculation, a common-time repaired-parent test at $t/\ell = 0.001$ established decreasing G9–G10–G11 discrepancies in every load-bearing field and physical-tensor sequence. The G10–G11 relative metric-increment and ADM- K discrepancies are respectively 2.82763×10^{-4} and 2.32536×10^{-4} , both below the fixed 2×10^{-3} ceiling; two complete G11 trajectories agree bitwise. This was field/tensor closure only, and the later direct G9/G10/G11 MOTS calculation at $t/\ell = 0.0015$ supplies the surface-observable result.

At $t/\ell = 0.001$, the fixed independent BVP detector accepts no branch from any of 12 seeds on any grid. The best profiles have independently evaluated positive outgoing expansion throughout their two-cell-interior regions. The subsequent same-parent continuation places this search before the critical time $t_*/\ell \simeq 0.00117254$ and the accepted pair at $t/\ell = 0.0015$. An earlier cross-mode acceleration comparison remains archived as an invalid same-mode spatial test; it is not used in the inference.

10 Negative, failed, and incomplete studies

Scientifically material negative and incomplete studies are preserved with their consequences:

- The initial no-root result is strong numerical evidence but not a formal interval proof. Consequently, the article states only bounded search nonacceptance.
- Formation times from the historical pre-repair larger-domain parents are not transferred to the repaired line or used to infer a universal formation threshold.
- The controlled post-fold interval supports a finite marginal-tube segment, not late-time settling, future completeness, or production evolution.
- Width and boundary variants that do not satisfy their own prerequisite or identity gates are not counted as adverse scientific outcomes.
- Arithmetic-precision and parent-qualification studies that preceded the canonical evolution establish their own bounded numerical conclusions; they do not retroactively convert this result into an event-horizon or continuum claim.

11 Reproducibility and archival mapping

The principal machine-readable aggregation is stored under the project archive identifier `Protocol 230` (archive-only saddle-node finalization, 24 August 2026) in `protocol230_result.json`. Its SHA-256 file hash is

`a54782417444dfe101b291a2b6d2594e5eb2e16f5774eab1500cf80328598156`.

It binds the immutable `Protocol 229` records on G9, G10, and G11. Their abbreviated SHA-256 hashes are, respectively, `a9ef9824...28b4e`, `4b3cb413...3dab`, and `1884b005...4f40`. The preceding direct formation closure is stored as `Protocol 228`, whose result file hash is `d82000ec...3f97`. `Protocol 226` remains the earlier field/tensor closure record rather than the primary scientific result. The existing-data precision and fit audit is stored as `Protocol 231`; its result SHA-256 is

`7e2f0f30ba6e0722e3f04c637557d97e837581ea7a6a2686ef6e520c21565b18`.

The independent G10 half-timestep evolution and continuation are stored as `Protocol 232`. Its result JSON and numerical archive have SHA-256 hashes

`16eea6a208b7db8be4a3cd090778e98b18f905f00ca32cba9d534e4cd25ea522`,
`98e53d892e4732398dea40b623517b43359f4e7b7f1d735286bd388797bbd999`.

The post-fold spatial-transfer, local-balance, integrated-balance, and half-timestep causal-signature records are stored as `Protocols 247--250`. Their result JSON SHA-256 hashes are, respectively,

`515e365bd7d8f1b1caeb7356724c0d3ca129364556de64898cf563cd8b122642`,
`66c938d71c7ad2a135d9f03dec48d1ebffcec6cd479ad031e48b2cd4a55f57a9`,
`6d54755023ce09beaa992e57a1eef1c1d2fcd3f42db0e662d08c25bd557e9317`,
`6866d08889beb5107436c3975b7d69ff7f48a80d244ff185ea954292ede400b3`.

These records preserve the separately evolved parents, the fixed leaf and segment selections, all three velocity paths and spatial stencils, the brane-inclusive ledger, recovery replays, and the explicit nonauthorization of event-horizon, topology, source-ownership, and continuum claims. The archival package binds source manifests, immutable inputs, environment receipts, prospective gate definitions, checkpoints, repeat traces, and final artifacts. The article and this supplement report scientific outcomes; the archive provides machine-level replay evidence. Machine-readable result records and frozen

protocol manifests are archived in the Zenodo version identified by [doi:10.5281/zenodo.22148554](https://doi.org/10.5281/zenodo.22148554). The complete version series is identified by the concept DOI [doi:10.5281/zenodo.22085054](https://doi.org/10.5281/zenodo.22085054). The version-controlled source code, tests, frozen protocol implementations, and figure-generation scripts are publicly available through the [GitHub repository](#) linked from that Zenodo record. Zenodo is the citable authority for the saddle-node and marginal-tube result records; GitHub is the public code repository.

12 Boundary of inference

The combined evidence does not establish:

- the absence of all marginal surfaces outside the bounded search classes;
- a measured continuum convergence order or extrapolated continuum value;
- a nonprincipal spectral-gap statement outside the $SO(3)$ -symmetric perturbation sector;
- nonlinear stability or a complete basin of attraction;
- an event horizon, common horizon, throat, parent/child universe, or topology change;
- a universal formation time or critical-amplitude law; or
- future-complete or late-time production evolution beyond the tested marginal-tube segment.

These questions require separate, prospectively defined studies. In particular, global event-horizon reconstruction would constitute a distinct paper rather than an appendix to the present quasi-local claim. The reflected quasi-local balance likewise does not assign mass to the visible, bulk, brane, or adjacent sector.

References

- [1] Abhay Ashtekar and Badri Krishnan. Dynamical horizons: Energy, angular momentum, fluxes, and balance laws. *Physical Review Letters*, 89:261101, 2002.
- [2] Abhay Ashtekar and Badri Krishnan. Dynamical horizons and their properties. *Physical Review D*, 68:104030, 2003.
- [3] Ivan Booth, Graham Cox, and Chiamaka Mary Okpala. Black hole evolutions: Lessons from bifurcation theory. *arXiv preprint*, 2026. arXiv:2605.30458, version 1.
- [4] Tetsuya Shiromizu and Masaru Shibata. Black holes in the brane world: Time symmetric initial data. *Physical Review D*, 62:127502, 2000.
- [5] Daoyan Wang and Matthew W. Choptuik. Black hole formation in Randall–Sundrum II braneworlds. *Physical Review Letters*, 117:011102, 2016.
- [6] Aghil Alaei, Martin Lesourd, and Shing-Tung Yau. Stable surfaces and free boundary marginally outer trapped surfaces. *Calculus of Variations and Partial Differential Equations*, 60:186, 2021.
- [7] Xiaoxiang Chai. A curvature estimate for stable marginally outer trapped hypersurface with a free boundary. *International Mathematics Research Notices*, 2024(6):4624–4655, 2024.
- [8] Abraão Mendes. Rigidity of free boundary MOTS. *Nonlinear Analysis*, 220:112841, 2022.
- [9] Sanghun Lee. Marginally outer trap surface with capillary boundary and rigidity of initial data sets. *arXiv preprint*, 2025. arXiv:2507.16617, version 1.
- [10] Deivid de Almeida and Abraão Mendes. Rigidity results for free boundary hypersurfaces in initial data sets with boundary. *Letters in Mathematical Physics*, 116:21, 2026.
- [11] Alexander Pigazzini. A degenerate free boundary minimal annulus and non-uniqueness in spherical caps beyond the hemisphere, 2026.

- [12] Lisa Randall and Raman Sundrum. A large mass hierarchy from a small extra dimension. *Physical Review Letters*, 83:3370–3373, 1999.
- [13] Walter D. Goldberger and Mark B. Wise. Modulus stabilization with bulk fields. *Physical Review Letters*, 83:4922–4925, 1999.
- [14] Oliver DeWolfe, Daniel Z. Freedman, Steven S. Gubser, and Andreas Karch. Modeling the fifth dimension with scalars and gravity. *Physical Review D*, 62:046008, 2000.
- [15] Norihiro Tanahashi and Takahiro Tanaka. Black holes in braneworld models. *Progress of Theoretical Physics Supplement*, 189:227–268, 2011.
- [16] Werner Israel. Singular hypersurfaces and thin shells in general relativity. *Il Nuovo Cimento B*, 44:1–14, 1966.
- [17] G. W. Gibbons and S. W. Hawking. Action integrals and partition functions in quantum gravity. *Physical Review D*, 15:2752–2756, 1977.
- [18] Richard A. Battye and Brandon Carter. Generic junction conditions in brane-world scenarios. *Physics Letters B*, 509:331–336, 2001.
- [19] Tim-Torben Paetz and Walter Simon. Marginally outer trapped surfaces in higher dimensions. *Classical and Quantum Gravity*, 30(23):235005, 2013.
- [20] Lars Andersson, Marc Mars, and Walter Simon. Local existence of dynamical and trapping horizons. *Physical Review Letters*, 95:111102, 2005.
- [21] Lars Andersson, Marc Mars, and Walter Simon. Stability of marginally outer trapped surfaces and existence of marginally outer trapped tubes. *Advances in Theoretical and Mathematical Physics*, 12(4):853–888, 2008.
- [22] Lars Andersson, Marc Mars, Jan Metzger, and Walter Simon. The time evolution of marginally trapped surfaces. *Classical and Quantum Gravity*, 26:085018, 2009.
- [23] Daniel Pook-Kolb, Ofek Birnholtz, Badri Krishnan, and Erik Schnetter. The existence and stability of marginally trapped surfaces. *Physical Review D*, 99:064005, 2019.
- [24] Frans Pretorius. Numerical relativity using a generalized harmonic decomposition. *Classical and Quantum Gravity*, 22:425–451, 2005.
- [25] Herbert B. Keller. Numerical solution of bifurcation and nonlinear eigenvalue problems. In Paul H. Rabinowitz, editor, *Applications of Bifurcation Theory*, volume 38 of *Publication of the Mathematics Research Center*, pages 359–384. Academic Press, New York, 1977.
- [26] Yuri A. Kuznetsov. *Elements of Applied Bifurcation Theory*, volume 112 of *Applied Mathematical Sciences*. Springer, New York, 3 edition, 2004.